

P100 Unit Diagnostic Report

Modem Failure - Sent for Repair

Node (planename)	N096AV
Airline	AVA (Avianca)
Serial Number	W15071881522
Modem Model	Sierra Wireless EM7565
Modem Firmware	SWI9X50C_01.14.02.00 (correct, not the root cause)
Diagnostic Date	August 20, 2026
Diagnosis	Hardware failure - SIM interface controller
Modem State	state: failed / failed reason: sim-missing

Diagnostic Summary

Unit N096AV has a Sierra Wireless EM7565 modem that fails to detect the SIM card, consistently reporting failed reason: sim-missing. The modem enumerates correctly over USB, firmware is at the correct version (SWI9X50C_01.14.02.00), and the operating system's serial ports/symlinks are healthy. Software causes were ruled out through the troubleshooting steps detailed below, leading to the conclusion that the failure is hardware-related (SIM interface controller).

Troubleshooting Steps Performed (in order)

1. Modem status check via ModemManager (mmcli -m 0): confirms the modem is detected over USB with correct firmware, but reports state 'failed' with reason 'sim-missing'.
2. Physical inspection: the SIM card was removed and inspected (no visible damage) and reseated firmly in its slot. The issue persisted with no change.
3. System symlink check (/dev/modemPPP, /dev/modemAT): both present and correctly pointing to their corresponding ttyUSB ports - rules out a udev rules or USB enumeration issue.
4. Attempted direct SIM status read via AT+CPIN: could not be executed (ModemManager blocks direct AT commands outside of debug mode), so the state reported by the QMI/ModemManager layer was used as reference.
5. Full modem reset (mmcli -m 0 --reset): after the reset and reinitialization, the modem reports the exact same state (failed / sim-missing) - rules out a stuck QMI session or a transient initialization issue.

Conclusion: with modem hardware, firmware, system symlinks, and the physical SIM ruled out as the cause through the steps above, and no recovery after a full modem reset, the failure is attributed to the SIM interface controller of the Sierra Wireless EM7565 modem. Modem replacement is recommended.

Appendix: Technical Evidence (logs and command output)

Unit Identification

N096AV
AVA
Serial Number: W15071881522
UUID: 03000200-0400-0500-0006-000700080009
Wake-up Type: Power Switch
SKU Number: SGP100-713020-00;M0-000;W0-100;B0-15
Date: Thu Aug 20 13:49:18 UTC 2026

Modem Status (mmcli -m 0) - before reset

General | path: /org/freedesktop/ModemManager1/Modem/0
| device id: 3a89be6bb0c26b97c0b3a074c595361c895519f9
Hardware | manufacturer: Sierra Wireless, Incorporated
| model: EM7565
| firmware revision: SWI9X50C_01.14.02.00 2e210b jenkins 2020/08/19 14:18:39
| h/w revision: 1.0
| equipment id: 353533103893122
System | device: /sys/devices/pci0000:00/0000:00:15.0/usb1/1-4/1-4.1
| drivers: qmi_wwan, qcserial
| ports: cdc-wdm0 (qmi), ttyUSB3 (qcdm), ttyUSB4 (gps), ttyUSB5 (at), wwan0 (net)
Status | state: failed
| failed reason: sim-missing
| power state: on
| signal quality: 0% (cached)
SIM | sim slot paths: slot 1: none (active)
| slot 2: none

Modem Status (mmcli -m 0) - after mmcli --reset

mmcli -m 0 --reset
successfully reseted the modem

Status | state: failed
| failed reason: sim-missing
| power state: on
SIM | sim slot paths: slot 1: none (active)
| slot 2: none

(No change from the pre-reset state)

System Symlinks

lrwxrwxrwx 1 root root 7 Aug 20 13:40 /dev/modemAT -> ttyUSB4
lrwxrwxrwx 1 root root 7 Aug 20 13:40 /dev/modemPPP -> ttyUSB5

udev Rules (20-modem-sierra-wireless.rules)

```
KERNELS=="*", SUBSYSTEMS=="usb", DRIVERS=="qcserial", ATTRS{bInterfaceNumber}=="01", SYMLINK+="modem-if-01"  
KERNELS=="*", SUBSYSTEMS=="usb", DRIVERS=="qcserial", ATTRS{bInterfaceNumber}=="03", SYMLINK+="modemPPP"  
KERNELS=="*", SUBSYSTEMS=="usb", DRIVERS=="qcserial", ATTRS{bInterfaceNumber}=="02", SYMLINK+="modemAT"
```

dmesg - Modem USB Enumeration

```
[ 139.370041] usb 1-4.1: new high-speed USB device number 7 using xhci_hcd
[ 139.471742] usb 1-4.1: New USB device found, idVendor=1199, idProduct=9091, bcdDevice=0.06
[ 139.471754] usb 1-4.1: Product: Sierra Wireless EM7565 Qualcomm Snapdragon X16 LTE-A
[ 139.471759] usb 1-4.1: Manufacturer: Sierra Wireless, Incorporated
[ 139.471763] usb 1-4.1: SerialNumber: UF22173591021052
[ 139.493555] qcserial 1-4.1:1.0: Qualcomm USB modem converter detected
[ 139.493694] usb 1-4.1: Qualcomm USB modem converter now attached to ttyUSB3
[ 139.494316] qcserial 1-4.1:1.2: Qualcomm USB modem converter detected
[ 139.494429] usb 1-4.1: Qualcomm USB modem converter now attached to ttyUSB4
[ 139.494962] qcserial 1-4.1:1.3: Qualcomm USB modem converter detected
[ 139.495157] usb 1-4.1: Qualcomm USB modem converter now attached to ttyUSB5
[ 139.501813] qmi_wwan 1-4.1:1.8: cdc-wdm0: USB WDM device
[ 139.502331] qmi_wwan 1-4.1:1.8 wwan0: register 'qmi_wwan' at usb-0000:00:15.0-4.1
```

AT+CPIN? (via mmcli --command)

```
error: command failed: 'GDBus.Error:org.freedesktop.ModemManager1.Error.Core.Unauthorized:
Cannot send AT command to modem: operation only allowed in debug mode'
```

Note: SIM PIN status could not be confirmed directly through this method;
SIM status was instead determined via the ModemManager/QMI report.

ModemManager - Relevant Initialization Logs

```
Aug 20 13:40:35 p100-ify ModemManager[810]: <info> [modem0] state changed (unknown -> locked)
Aug 20 13:40:35 p100-ify ModemManager[810]: <warn> [modem0] modem couldn't be initialized:
Couldn't check unlock status: QMI operation failed: GW primary session index unknown
Aug 20 13:40:35 p100-ify ModemManager[810]: <info> [modem0] state changed (locked -> failed)
```

Note: a full modem reset (mmcli -m 0 --reset) was performed after this log,
with no change in outcome - rules out a stuck QMI session as a transient cause.